11 In this question you must show detailed reasoning.

The variables x and y are such that $\frac{dy}{dx}$ is directly proportional to the square root of x.

When $x = 4$, $\frac{dy}{dx} = 3$.

(a) Find $\frac{dy}{dx}$ in terms of x. **[3]**

When $x = 4$, $y = 10$.

(b) Find y in terms of x. **[3]**